41) Let f be a function from A to B. Let S be a subset of B. Show that $f^{-1}(\overline{S}) = \overline{f^{-1}(S)}$.

Soln: Given, $f: A \to B$, $S \subseteq B$.

Let, $x \in f^{-1}(\overline{S})$. By defn. of inverse image, $f(x) \in \overline{S}$. By Defn. 5 from Sec-2.2, $f(x) \in U$, $f(x) \notin S$. By defn. of not belongs to, $\neg(f(x) \in S)$ is True. By Defn. of negation, $\neg(\neg(f(x) \in S))$ is False. By Double-negation Law of propositions, $f(x) \in S$ is False. By defn. of inverse image, $x \in f^{-1}(S)$ is False. By defn. of negation, $\neg(x \in f^{-1}(S))$ is True. By ~~Double negation law of propositions~~, Defn. of not belongs to, $x \notin f^{-1}(S)$ is True. $\because x \in U$, By Defn. 5 from Sec-2.2, $x \in \overline{f^{-1}(S)}$. By Defn. 3 from Sec-2.1, $f^{-1}(\overline{S}) \subseteq \overline{f^{-1}(S)}$.

Let, $x \in \overline{f^{-1}(S)}$. By Defn. 5 from Sec-2.2, $x \in U$, $x \notin f^{-1}(S)$. By Defn. of not belongs to, $\neg(x \in f^{-1}(S))$ is True. By Defn. of negation, $\neg(\neg(x \in f^{-1}(S)))$ is False. By Double-Negation Law of Propositions, $x \in f^{-1}(S)$ is False. By Defn. of inverse image, $f(x) \in S$ is False. By Defn. of negation, $\neg(f(x) \in S)$ is True. By Defn. of not belongs to, $f(x) \notin S$ is True. $\because f(x) \in U$, By Defn. 5 from Sec-2.2, $f(x) \in \overline{S}$. By Defn. of inverse image, $x \in f^{-1}(\overline{S})$. By Defn. 3 from Sec-2.1, $\overline{f^{-1}(S)} \subseteq f^{-1}(\overline{S})$.

$\therefore f^{-1}(\overline{S}) = \overline{f^{-1}(S)}$ □

50) Show that if x is a real number, then $\lceil x \rceil - \lfloor x \rfloor = 1$ if x is not an integer and $\lceil x \rceil - \lfloor x \rfloor = 0$ if x is an integer.

Soln: Let, $\lfloor x \rfloor = m$ and $\lceil x \rceil = n$. By Defn. 12, m and n are integers. Property (1a) gives $m \leq x < m+1$. Property (1b) gives $n-1 < x \leq n$. The smallest integer $\geq x$ is n. The largest integer $\leq x$ is m.

Suppose, x is not an integer.

$\therefore m < x < m+1$ and $n-1 < x < n$. (Wrong)